

Data

12 frozen encoders

Stage 1 analyses

mRNA stability
(CodonBERT, n = 981)

SAME input:
one coding sequence read
as DNA / RNA / protein

GTEX expression
(IsoFormer, n = 999 + 996)

DISTINCT inputs:
TSS DNA window / transcript
/ protein; 30 tissues

DNA	RNA	Protein
NT 500M	RNA-FM	ESM-2 8M
NT v2 100M	RiNALMo	ESM-2 35M
DNABERT-2	mRNA-FM*	ESM-2 150M
HyenaDNA	CaLM*	ProtBERT

* *codon tokens* | *mean-pooled final layer*

Track A: Decodability

linear probes: stability, expression,
GC / GC3, length; concatenation gains

Track B: Geometry

CKA, k-NN, CCA retrieval, RSA;
before / after composition control

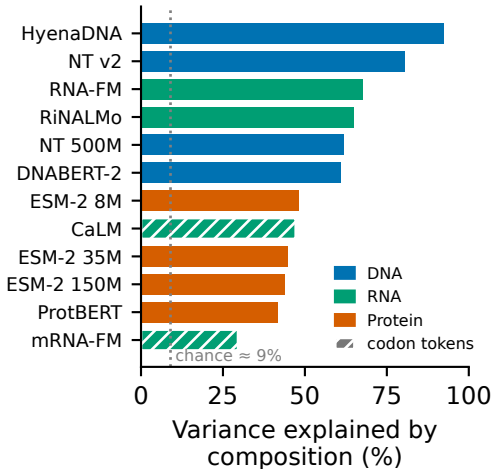
Track C: Diagnostics

layer-wise CKA, attention vs motifs,
per-sequence alignment vs error

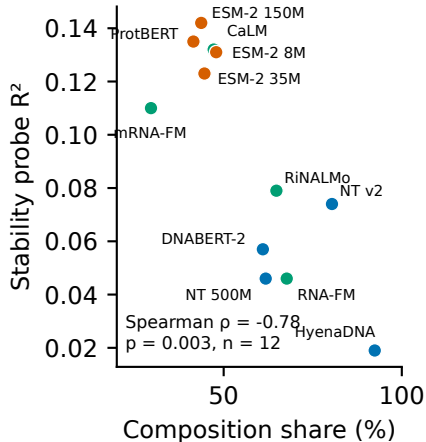
Stage 2: define alignment
alignment = held-out agreement AFTER
the composition control, reported separately
from predictive complementarity

Stage 3: fusion guided by measurement
no alignment loss by default; codon grid only for
same-input data; codon-RNA + protein first;
must beat length, concatenation, best single encoder

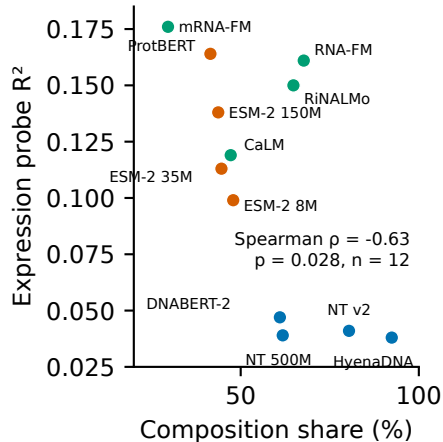
a Composition share



b mRNA stability

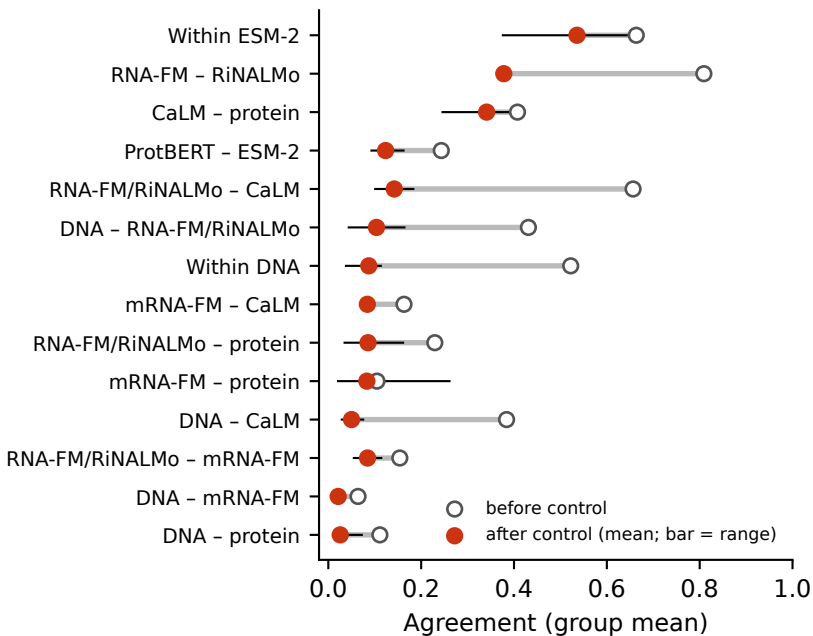


c GTEx expression

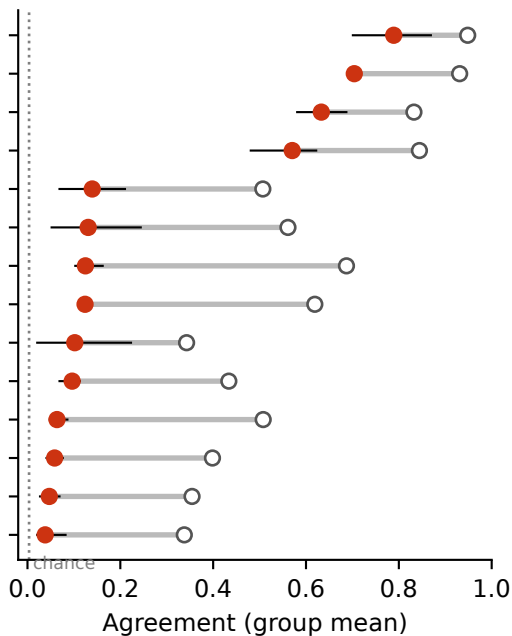


Agreement between encoders before and after removing composition (stability data, 66 pairs)

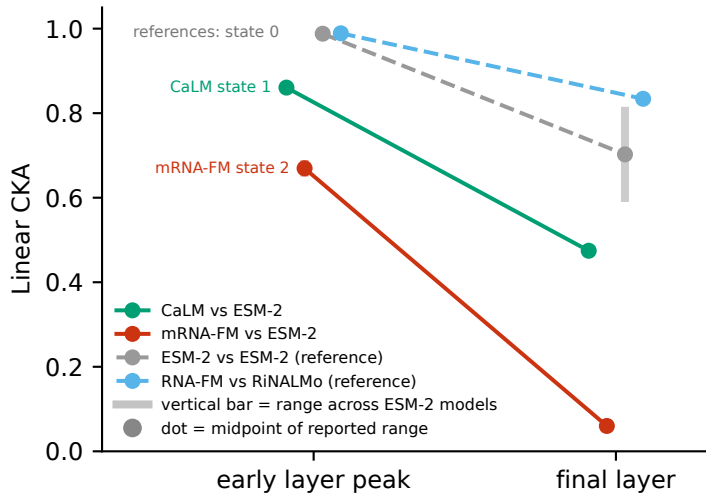
a Linear CKA



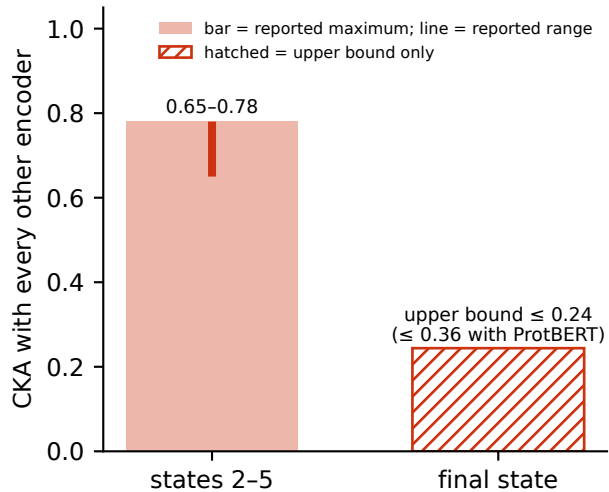
b Retrieval Recall@1



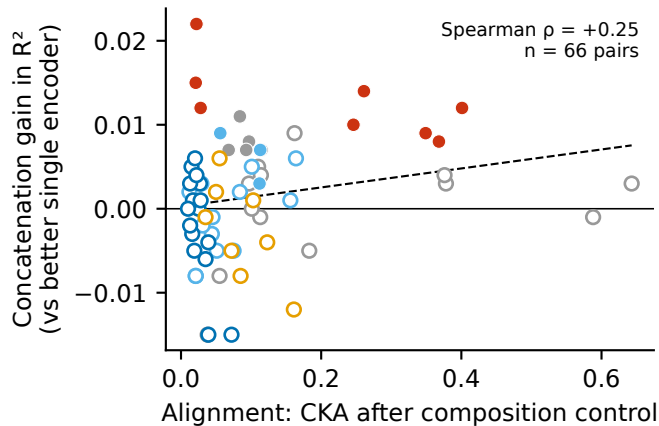
a Where agreement lives



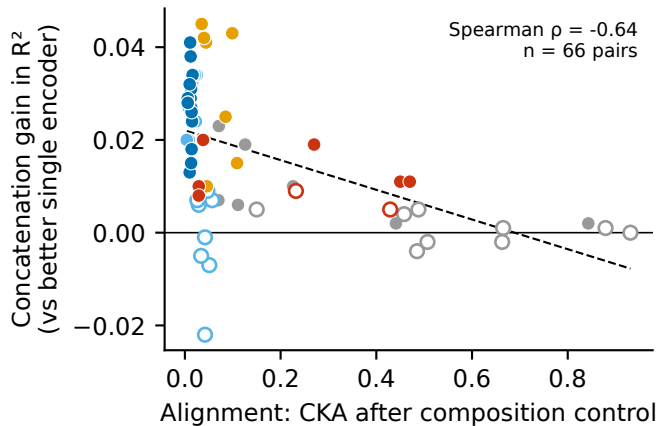
b mRNA-FM diverges in its last blocks



a mRNA stability: same input



b GTEx: distinct inputs



Colour: pair type

● within modality
 ● DNA - RNA
 ● DNA - protein
 ● nucleotide RNA - protein
 ● codon RNA - protein

Marker

● filled = gain under all 10 fold assignments (consistent)
 ○ open = gain under fewer than 10 assignments (inconsistent)
 --- least-squares trend